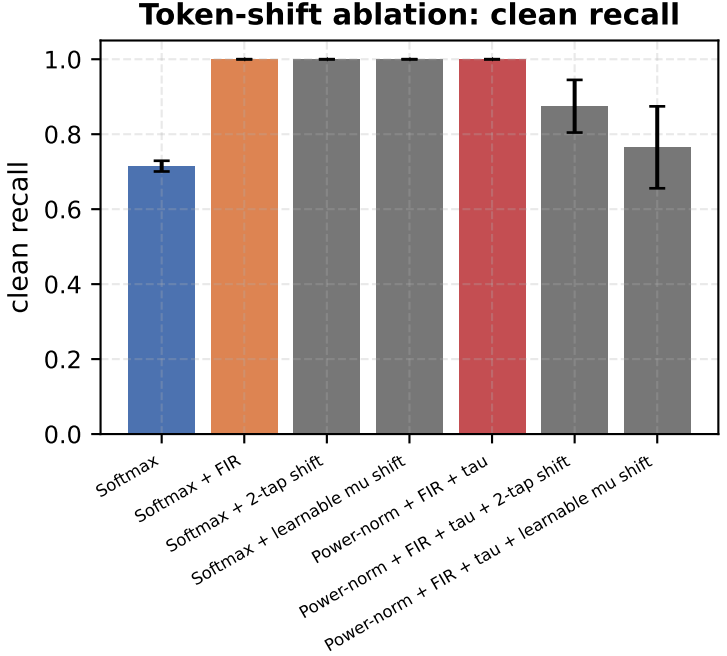
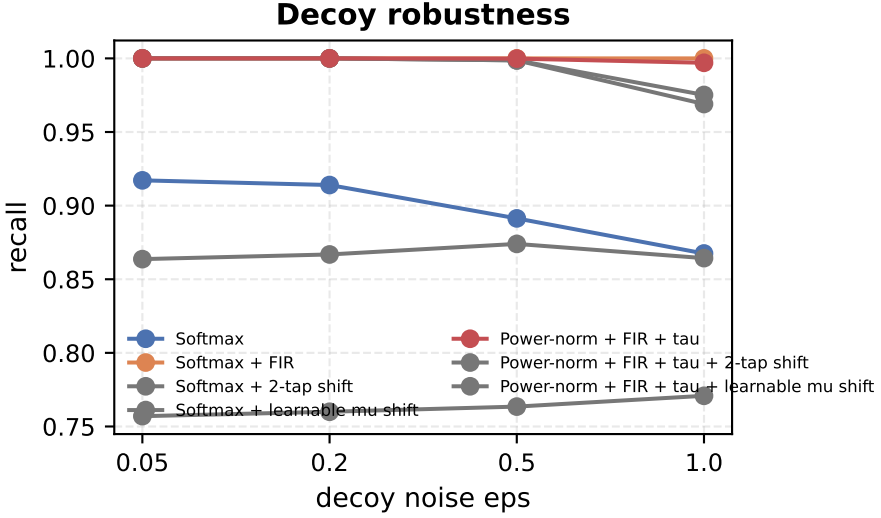
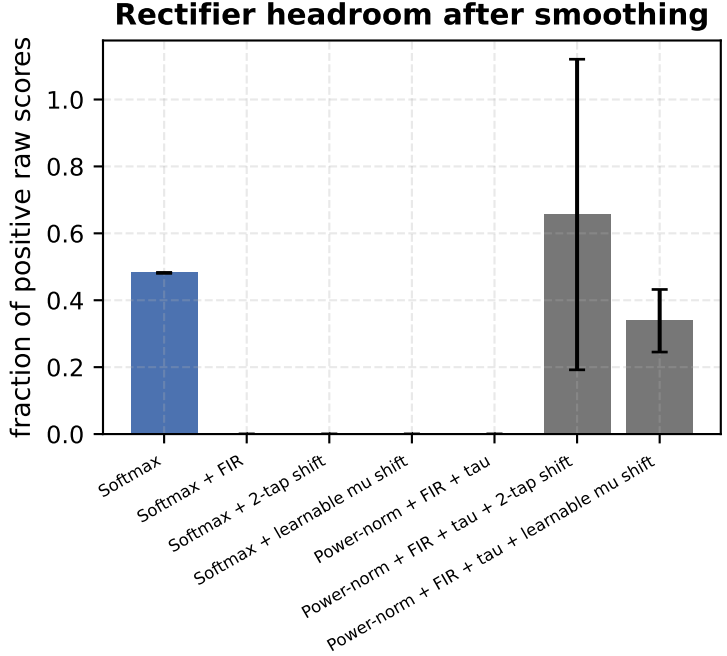

Token-shift ablation: clean recall
clean recall
1.0
0.8
0.6
0.4
0.2
0.0
Softmax
Softmax + FIR
Softmax + 2-tap shift
Softmax + learnable mu shift
Power-norm + FIR + tau
Power-norm + FIR + tau + 2-tap shift
Power-norm + FIR + tau + learnable mu shift


Decoy robustness
recall
1.00
0.95
0.90
0.85
0.80
0.75
0.05
0.2
0.5
1.0
decoy noise eps
Softmax
Softmax + FIR
Softmax + 2-tap shift
Softmax + learnable mu shift
Power-norm + FIR + tau
Power-norm + FIR + tau + 2-tap shift
Power-norm + FIR + tau + learnable mu shift


Rectifier headroom after smoothing
fraction of positive raw scores
1.0
0.8
0.6
0.4
0.2
0.0
Softmax
Softmax + FIR
Softmax + 2-tap shift
Softmax + learnable mu shift
Power-norm + FIR + tau
Power-norm + FIR + tau + 2-tap shift
Power-norm + FIR + tau + learnable mu shift